

FDDT Case Report

Fraud Document Detection Tool — system-assisted analysis for human review. See Section 6 for what this report is and is not.

1. Case Details

Case ID	CASE-3CCC350B 3ccc350b-6704-4613-8cea-7aeffedf0dc9
Submitted by	testreviewer L!
Submission date	2026-10-07 06:44 UTC
Documents	1 Types: School document
Case type	School document
Status / decision	Pending manual review / Pending review
Report generated	2026-10-07 06:52 UTC by testreviewer L! (Reviewer L1)
Report ID	98627617-cd42-4f0a-b535-84391031ee79

2. Executive Summary

MEDIUM RISK **Score 37 / 100** (computed 2026-10-07 06:50 UTC)

Why: Driven mainly by (15 points): 'case12School educational document.pdf': Tax 0.00 is not 5% of the subtotal 70,360.00 (expected 3,518.00). No combination of the line items explains it either.

Decision status: Pending review.

Exceptions: 7 item(s) did not pass, 7 of them shown as highlighted regions on the source pages. Section 5 lists each one; Section 9 shows them on the documents.

This summary is deliberately short. The reasons behind the tier are in Section 3, the complete checklist is in Section 4.

3. Explainable Findings

The risk engine is a transparent, weighted rule set. Each row is one rule that fired, with the reason text stored at scoring time. Points are summed (raw 37), capped at 100, and mapped to a tier (medium from 30, high from 60) — here **37/100, medium**.

Severity	Points	Check	Document / page	Reason
MEDIUM	15	Field validation	case12School educational document.pdf · page 1	'case12School educational document.pdf': Tax 0.00 is not 5% of the subtotal 70,360.00 (expected 3,518.00). No combination of the line items explains it either. Highlights: R6, R7
MEDIUM	12	Visual inconsistency / AI-generation review	case12School educational document.pdf · page 1	'case12School educational document.pdf': the vision-model review reports inconsistent fonts. Page 1 — Font consistency: The invoice number and invoice date fields use a bold font in a blue background, which differs from the plain text font used in the rest of the document, such as the item descriptions and amounts. This is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a definitive finding. Highlights: R5
MEDIUM	10	Metadata forensics	case12School educational document.pdf	'case12School educational document.pdf' was last saved by a PDF editor, not a scanner or billing system. The file was last written by a PDF editor ('iLovePDF' in info./Producer), not by a scanner or the issuer's billing system — the tool can change the text and numbers of an existing PDF.

4. All Checks

The completeness record: every check that ran on every document, including those that passed. Section 3 is the narrative; this is the checklist.

Document 1 — case12School educational document.pdf (School document · 4 passed, 4 flagged)

Check	Result	Details
Metadata forensics PDF metadata, edit history and incremental-save markers.	Flag	1 finding(s). The file was last written by a PDF editor ('iLovePDF' in info./Producer), not by a scanner or the issuer's billing system — the tool can change the text and numbers of an existing PDF.
Font consistency Text set in a different font family from the text around it (text layer, or OCR font recognition on scans).	Flag	4 finding(s). Page 1 (scanned): '@hotmail.com' looks like a different font (grotesque sans) from the text around it (serif), per OCR font recognition — it may have been typed in or pasted before the page was printed or scanned. Only 2 word(s) on this page loo...
Error level analysis (ELA) Pixel-level recompression differences that can indicate edited regions.	Pass	No anomaly detected by this check.
Copy-move detection Regions of a page that were duplicated elsewhere on the same page.	Pass	No anomaly detected by this check.
Visual inconsistency / AI-generation review Model review of fonts, alignment, contrast, sharpness, lighting and signs of AI generation.	Flag	1 finding(s). Page 1 — Font consistency: The invoice number and invoice date fields use a bold font in a blue background, which differs from the plain text font used in the rest of the document, such as the item descriptions and amounts. This is a vision-mode...
Signature / stamp detection Whether a signature or stamp mark is present where this kind of document normally carries one.	Pass	2 signature/stamp region(s) located (presence only).
Duplicate detection Perceptual-hash comparison of each page against previously submitted pages.	Pass	No anomaly detected by this check.
Field validation Date, subtotal / tax / total and reference-number plausibility.	Flag	Tax 0.00 is not 5% of the subtotal 70,360.00 (expected 3,518.00). No combination of the line items explains it either.
Issuer verification Issuer name matched against the issuer registry.	Not checked	Not checked: the issuer registry has no active entries for school issuers in AE, so there was nothing to compare this issuer against.

5. Exceptions

Everything that did **not** pass, drawn from Section 4 and written as concrete items (7 in total). A region ID (R1, R2, ...) points to the highlight that shows the exception on the source page in Section 9 — click it to jump there.

No.	Exception	Where	Highlight
1	Metadata forensics: The file was last written by a PDF editor ('iLovePDF' in info./Producer), not by a scanner or the issuer's billing system — the tool can change the text and numbers of an existing PDF.	Document 1 (case12School educational document.pdf)	—
2	Font consistency: text set in a different font family from the text around it (OCR font recognition on a scan) Page 1 (scanned): '@hotmail.com' looks like a different font (grotesque sans) from the text around it (serif), per OCR font recognition — it may have been typed in or pasted before the page was printed or scanned. Only 2 word(s) on this page look like this font, so it is shown for review but not scored: OCR font recog...	Document 1 (case12School educational document.pdf) · page 1	R1
3	Font consistency: text set in a different font family from the text around it (OCR font recognition on a scan) Page 1 (scanned): '5' looks like a different font (humanist sans) from the text around it (serif), per OCR font recognition — it may have been typed in or pasted before the page was printed or scanned. Only 1 word(s) on this page look like this font, so it is shown for review but not scored: OCR font recognition alone...	Document 1 (case12School educational document.pdf) · page 1	R2
4	Font consistency: text set in a different font family from the text around it (OCR font recognition on a scan) Page 1 (scanned): '2' looks like a different font (monospace) from the text around it (serif), per OCR font recognition — it may have been typed in or pasted before the page was printed or scanned. Only 1 word(s) on this page look like this font, so it is shown for review but not scored: OCR font recognition alone is...	Document 1 (case12School educational document.pdf) · page 1	R3
5	Font consistency: text set in a different font family from the text around it (OCR font recognition on a scan) Page 1 (scanned): 'emokhtar@brightoncollegealain.ae' looks like a different font (grotesque sans) from the text around it (serif), per OCR font recognition — it may have been typed in or pasted before the page was printed or scanned. Only 2 word(s) on this page look like this font, so it is shown for review but not sc...	Document 1 (case12School educational document.pdf) · page 1	R4
6	Visual review: font consistency inconsistency (model judgment) Page 1 — Font consistency: The invoice number and invoice date fields use a bold font in a blue background, which differs from the plain text font used in the rest of the document, such as the item descriptions and amounts. This is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a defin...	Document 1 (case12School educational document.pdf) · page 1	R5
7	Tax / rate miscalculation: Tax 0.00 is not 5% of the subtotal 70,360.00 (expected 3,518.00). No combination of the line items explains it either.	Document 1 (case12School educational document.pdf) · page 1	R6, R7

1 exception(s) show “—” because they have no place on a page (for example a missing field, or a check that did not localize its finding).

6. Limitations and Assumptions

Please read this section before relying on anything in this report.

Limitations of This Analysis

What this report is

This is **system-assisted analysis to support human review**. It is not a certified forensic or legal determination that any document is genuine, forged or fraudulent, and it must not be presented as one.

Three kinds of finding

Deterministic pixel and structure checks — error level analysis, copy-move detection and PDF metadata forensics — are computed by fixed algorithms over the file’s pixels and structure. Localized findings are drawn with **solid** boxes (ELA in red, copy-move in orange). They are signals, not proof: ELA in particular can false-positive on legitimately recompressed scans and can miss print-and-rescan tampering.

Deterministic rule-based field checks — field validation and cross-document consistency — compare extracted values by fixed rules and are drawn with **solid purple** boxes. They are exact about the values they compare, but only as reliable as the OCR and extraction that produced those values.

Model-judgment checks — the visual inconsistency review, the AI-generation assessment and the signature/stamp comparison — are **probabilistic assessments** made by a vision-language model, not deterministic proof. They can vary from run to run, and their boxes are only approximate; they are the **dashed** boxes captioned “approximate” throughout this report. Signature and stamp detection shows presence and placement only and is not an identity match; the AI-generation assessment is experimental, and its accuracy on scanned business documents has not been validated.

What the system can and cannot see

The system evaluates the **submitted document file itself**. It cannot verify facts external to the document — for example whether a real-world transaction actually occurred, whether the named issuer really produced the document, or (without a pre-existing reference signature) who a signatory is. Printed Arabic and English text are read; handwritten Arabic is not reliably read.

Absence of a finding

The absence of a flagged finding does not guarantee that a document is genuine. It means only that no supported check detected an anomaly.

Check coverage on this case

Not every check applies to every document or case. This is what actually ran here:

Check	Status	Coverage on this case
Metadata forensics	Ran	Ran on 1 of 1 document(s); 1 flagged.
Font consistency	Ran	Ran on 1 of 1 document(s); 1 flagged.
Error level analysis (ELA)	Ran	Ran on 1 of 1 document(s); 0 flagged.
Copy-move detection	Ran	Ran on 1 of 1 document(s); 0 flagged.
Visual inconsistency / AI-generation review	Ran	Ran on 1 of 1 document(s); 1 flagged.
Signature / stamp detection	Ran	Ran on 1 of 1 document(s); 0 flagged.
Duplicate detection	Ran	Ran on 1 of 1 document(s); 0 flagged.
Field validation	Ran	Ran on 1 of 1 document(s); 1 flagged.
Issuer verification	Ran	Ran on 1 of 1 document(s); 0 flagged.
Signature comparison (advisory)	Did not run	Did not run — no reference signature was set for this case.
Cross-document consistency	Not applicable	Not applicable — this case has a single document.

Risk score and rule versions

The risk score in this report (**37/100, medium**) was computed at 2026-10-07 06:50 UTC (assessment 2b39edf2-1491-4df8-8e54-ff31d1f5a286) from the risk-rule versions that fired: `field.tax_rate_mismatch v2` (weight 15); `metadata.pdf_editor_producer v1` (weight 10); `visual.font_inconsistency v1` (weight 12). Tier thresholds at that time:

medium from 30, high from 60.

Rule weights are versioned and are never re-applied retroactively: if an administrator later edits a rule, this case is not re-scored, and **this report remains an accurate historical record** of what was known and weighted when it was generated. A report generated later may legitimately differ.

Assumptions

What the analysis took to be true without independently verifying it. These are specific to what ran on this case.

- The issuer registry (0 active entries) was assumed to be current and accurate as of 2026-10-07 06:52 UTC. An issuer that is absent from it is reported as “not in registry”, which is a prompt to verify the issuer — not a finding that the issuer is fraudulent.
- Field values were read by OCR followed by language-model extraction and are assumed to have been read correctly. Each field carries a confidence and an “uncertain” flag. This build records that flag but does not automatically route low-confidence documents to manual review, so it is listed here for the reviewer instead: no field was marked uncertain on this case.
- Purple field highlights were located by matching extracted values back onto OCR word positions; treat their position as approximate. The comparison behind each one uses the stored, normalized values, not the highlight.
- The SHA-256 chain of custody shows the stored file is unchanged since upload. It does not show that the uploaded file is the issuer’s original — a document altered before upload is judged on its content, like any other.
- Risk-rule weights, severities and tier thresholds are administrator-configured and assumed to be appropriately calibrated; the score is advisory and is not a probability of fraud.

7. Audit Trail

Chronological case-level events, read from the append-only audit log (times in UTC). This report does not write to or duplicate the log.

Time	Event	Actor	Detail
2026-10-07 06:44 UTC	Case created	testreviewer L! (Reviewer L1)	
2026-10-07 06:44 UTC	Document uploaded	testreviewer L! (Reviewer L1)	case12School educational document.pdf; SHA-256 5ba396aed871b103...
2026-10-07 06:47 UTC	Font consistency check completed	System	case12School educational document.pdf
2026-10-07 06:47 UTC	Text extraction, classification and field extraction completed	System	case12School educational document.pdf
2026-10-07 06:48 UTC	Visual inconsistency review completed	System	case12School educational document.pdf
2026-10-07 06:49 UTC	Signature / stamp detection completed	System	case12School educational document.pdf
2026-10-07 06:49 UTC	Field validation and issuer verification completed	System	case12School educational document.pdf
2026-10-07 06:49 UTC	Metadata forensics completed	System	case12School educational document.pdf
2026-10-07 06:49 UTC	Duplicate detection completed	System	case12School educational document.pdf
2026-10-07 06:49 UTC	ELA and copy-move checks completed	System	case12School educational document.pdf
2026-10-07 06:50 UTC	Risk score computed	System	tier medium, score 37
2026-10-07 06:50 UTC	Case status changed	System	submitted → pending_manual_review; automated checks complete

8. Appendix

The detailed technical record, for someone auditing the analysis itself rather than a first-pass reviewer.

8.1 Extracted fields

Normalized values as stored at extraction time. “Located” is the page the value was matched to (used for the purple highlights); a blank means no position was recorded. Arabic values are shown right-to-left, numbers left-to-right.

Document 1 — case12School educational document.pdf

Field	Value	Confidence	Located
Issuer	Bloom Academy LLC	95%	p.1
Reference number	BCAA054-24-25	95%	p.1
Date	2024-07-11	95%	p.1
Amount	70,360.00 AED	95%	p.1
Subtotal	70,360.00 AED	95%	p.1
Tax rate	5.00	90%	p.1
Tax amount	0.00 AED	95%	p.1
Trn number	100570428100017	95%	
Bill to	Abdullah Almazrouei	95%	p.1
Email	abdullahalmazrouei@hotmail.com	95%	p.1
Pupil id	BCAA101270	95%	p.1
Pupil name	Ahmed Abdullah Almazrouei	95%	p.1
Academic year	2024-2025	95%	p.1
Grade	FS5	95%	p.1
Discount	0.00	95%	p.1
Paid amount	0.00	95%	p.1
Net invoice amount	70,360.00	95%	p.1
Payment instructions	Please make payment by a cheque payable to BLOOM ACADEMY LLC Or By Bank Transfer To : Account Name: BLOOM ACADEMY LLC Bank Name: Dubai Islamic Bank Bank Address: P.O. BOX 3863, Abu Dhabi, U.A.E Account Number : 010-103-1203148-01 IBAN No: AE610240010103120314801 SWIFT CODE: DUIBAEAD	95%	
Contact info	Finance Department emokhtar@brightoncollegealain.ae Telephone: +971-3-7133999 or +971-3-7133917 Brighton College , Al Ain, U. A. E. P.O.Box: 101110, Al Ain Zakher Street	95%	

8.2 Document hashes (chain of custody)

SHA-256 of each original as recorded when it was uploaded to immutable storage, re-checked against the stored file when this report was generated.

Document	SHA-256	Verification
case12School educational document.pdf 118,065 bytes	5ba396aed871b103409779eab1a736d2802421f23fc1ee6d9175b5c54e4e15ea	VERIFIED

8.3 Technical parameters used by each check

Page rendering (input to ELA, copy-move, duplicate, visual review)

Render resolution	200 DPI
Colour model	RGB, converted to BGR for OpenCV

Metadata forensics

Libraries	pikepdf / PyMuPDF (PDF structure, XMP, incremental updates)
Modified-after-created rule	flagged if ModDate is more than 1 second(s) after CreationDate
Editing-software names matched	photoshop, gimp, illustrator, affinity, coreldraw, paint.net, pixlr, canva, inkscape

Error level analysis (ELA)

JPEG recompression quality	75
Error gain (scale)	50 (sqrt error term; the reference tool's default)
Contrast stretch / LUT	not applied — the raw error magnitude is thresholded directly
Region threshold	page-relative percentile 99.7 with an absolute floor of 60/255
Region area accepted	0.2% – 60% of the page
Applies to	pages with an embedded raster image only (born-digital text pages are exempt)

Copy-move detection

Keypoint algorithm	BRISK (OpenCV), detector threshold 200
Response / matching / distance thresholds	90 / 20 / 15
Minimum cluster size	30 matching keypoint pairs
Minimum region area	2% of the page
Keypoint / match caps	4,000 / 4,000

Duplicate detection

Hash	perceptual hash (imagehash pHash), 64-bit, one per rendered page
Hamming-distance threshold	5 (≤ this is flagged)
Compared against	every page hash of every previously submitted document

Visual inconsistency / AI-generation review

Model	gpt-4.1-mini
Independent runs per page	2
A finding is kept if	both runs report it, or one reports it with high confidence
Image sent to model	page render, long edge ≤ 1536px

Signature / stamp detection and comparison

Model	gpt-4.1-mini
Pages analysed per document	first 10
Output	presence and placement only; comparison is a qualitative verdict, never a numeric score

Field validation

Sub-checks	date not in the future; subtotal vs line items; total vs subtotal + tax; reference-number format
Amount tolerance	1% of the amount, minimum 0.01

Issuer verification

Matching	rapidfuzz weighted ratio against the active issuer registry, with a model-judged fallback for cross-script names
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Similarity threshold	85
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Field locations (purple highlights)

Method	extracted values matched back onto OCR word positions (Azure Document Intelligence Layout)
Repeated values	resolved by nearby keywords (“total”, “subtotal”, “VAT”, “date”...)
Accuracy	approximate; a field that cannot be matched is reported without a highlight

9. PDF Highlighted Regions

Every flagged page, rendered from the original with its findings drawn in. These are copies made for this report; the original files are untouched. Only pages with at least one highlight appear. Each highlight has a region ID that Section 5 refers back to.

Colour key

■	ELA (solid red)
■	Copy-move (solid orange)
■	Font mismatch in the text layer (solid fuchsia)
■	Field exception (solid purple)
■	Model judgment (dashed blue, approximate)
■	Page-level finding (not localized)

Region index

Region	Document	Page	What it shows
R1	Document 1 (case12School educational document.pdf)	1	Font mismatch — text set in a different font family (scan, estimated)
R2	Document 1 (case12School educational document.pdf)	1	Font mismatch — text set in a different font family (scan, estimated)
R3	Document 1 (case12School educational document.pdf)	1	Font mismatch — text set in a different font family (scan, estimated)
R4	Document 1 (case12School educational document.pdf)	1	Font mismatch — text set in a different font family (scan, estimated)
R5	Document 1 (case12School educational document.pdf)	1	Visual review — font consistency (approximate)
R6	Document 1 (case12School educational document.pdf)	1	Field exception: Tax amount (0.00 AED vs 5% = 3,518.00)
R7	Document 1 (case12School educational document.pdf)	1	Field exception: Tax rate (5% gives 3,518.00)

Section 9 · Document 1 of 1 · regions R1, R2, R3, R4, R5, R6, R7

Bloom Academy LLC



BRIGHTON COLLEGE
AL AIN

TAX INVOICE

TIN: 50570429100017

Bill To:

Email: abdul@bloomacademy.ae

Pupil ID: BCAA01270

Academic Year: 2024-2025

Grade: PS5

Item 5: Term 1 Transport fee

Item 6: Term 2 Transport fee

Item 7: Term 3 Transport fee

Item	Description	Unit Price	Quantity	Total
5	Term 1 Transport fee	3,518.00	1	3,518.00
6	Term 2 Transport fee	3,518.00	1	3,518.00
7	Term 3 Transport fee	3,518.00	1	3,518.00
Subtotal				10,554.00

AED Seventy Thousand

Subtotal: 10,554.00

Tax (5%): 527.70

Total: 11,081.70

AED Seventy Thousand

Field exception: Tax rate (5% gives 3,518.00)

Field exception: Tax amount (0.00 AED vs 5% = 3,518.00)

Payment Information:

Bank Name: BLOOM ACADEMY LLC

Bank Address: Dubai Islamic Bank

Account Number: 010-103-1203148-01

IBAN No: AE610240010103120314801

SWIFT CODE: DUBIAEAD

Signature: _____



Finance Department

Payment Information:

Bank Name: BLOOM ACADEMY LLC

Bank Address: Dubai Islamic Bank

Account Number: 010-103-1203148-01

IBAN No: AE610240010103120314801

SWIFT CODE: DUBIAEAD

AED Seventy Thousand

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AED Seventy Thousand

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AED Seventy Thousand

Field exception: Tax rate (5% gives 3,518.00)

Field exception: Tax amount (0.00 AED vs 5% = 3,518.00)

AED Seventy Thousand

Field exception: Tax rate (5% gives 3,51